

Surface potential measurements of a polymer film following primary ion gun irradiation for ToF-SIMS analysis of insulator using a Kelvin probe and the observation of effects from the vacuum gauge



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ARTICLE INFO

Keywords:

Surface charge neutralization
Surface potential decay
Ion beam
ToF-SIMS

ABSTRACT

When an insulator sample is analyzed via time-of-flight secondary ion mass spectrometry (ToF-SIMS), charges accumulate on the sample surface due to the primary ion beam. These local surface charges and surface potentials cause many problems during analysis. A Kelvin probe was used to measure the surface potentials of the insulator sample. The surface potential induced on the surface of the sample and its temporal change was measured following the irradiation of the gas cluster ion beam. It was confirmed that the estimated surface potential value is affected by the high vacuum gauge, and there is an inflow of charges that possess polarities opposite to that of the surface potential. The amount of current flowing from the high vacuum gauge was calculated using the surface potential data that was measured. The amount of charge flowing onto the surface of the insulator was varied from several tens of fA to several fA, which was depended on its vacuum degree and position of high vacuum ion gauge. Additionally, neutralization of surface charges by the neutral gas effect was also weakly observed. This leaked current from the high vacuum gauges is likely to have a small but lasting impact on the ToF-SIMS insulator analysis.

1. Introduction

Since the 1970s, time-of-flight secondary ion mass spectroscopic (ToF-SIMS) analysis has become a universal technology that is applied to the study of various organic materials such as polymers. Because it has high spatial resolution and capability for label-free chemical analysis. This technique has recently been expanded to permit bio-sample analysis using a cluster ion beam [1–3]. Over the past several years, the Korea Basic Science Institute (KBSI) has been developing cluster ion guns and ToF-SIMS for the 3D imaging analysis of organic materials and bio-samples [4,5].

Recently, many samples are bio-organic materials, and which are a kind of insulator. When such an insulator sample is analyzed via ToF-SIMS, charges accumulate on the sample surface due to the primary ion beam. The formation of these local surface charges and their potentials field causes many problems during analysis [6,7]. These surface potentials may result in the reduction or disappearance of the ion signal. As a method of controlling the surface potential, conventional methods such as irradiation with a low energy electron beam or ion beams and coating the surface with a conductive film have mainly been used

[6–9]. In particular, low-energy electron beam irradiation is commonly used for charge neutralization [10]. However, the proper conditions under which this electron beam may be applied to the sample surface depend on the sample itself, therefore requiring careful control methods. Additionally, the chemical properties of a sample may be damaged via irradiation with an electron beam [11]. Therefore, the use of electron beam irradiation when analyzing an insulator sample should be carefully controlled.

In this paper, a gas cluster ion beam (GCIB) was irradiated onto the surface of a polymer insulator sample to accumulate local charges on the surface. A Kelvin probe was used to measure the temporal changes of the surface potential. Using these measurements of a Kelvin probe, the influence of the high vacuum gauge and neutral gas on the sample surface and surface potential will be determined.

2. Experimental methods

Fig. 1 shows the surface potential measurement system used in this study, which consists of four components: The first component is the GCIB gun that irradiates a primary ion beam onto the sample surface.

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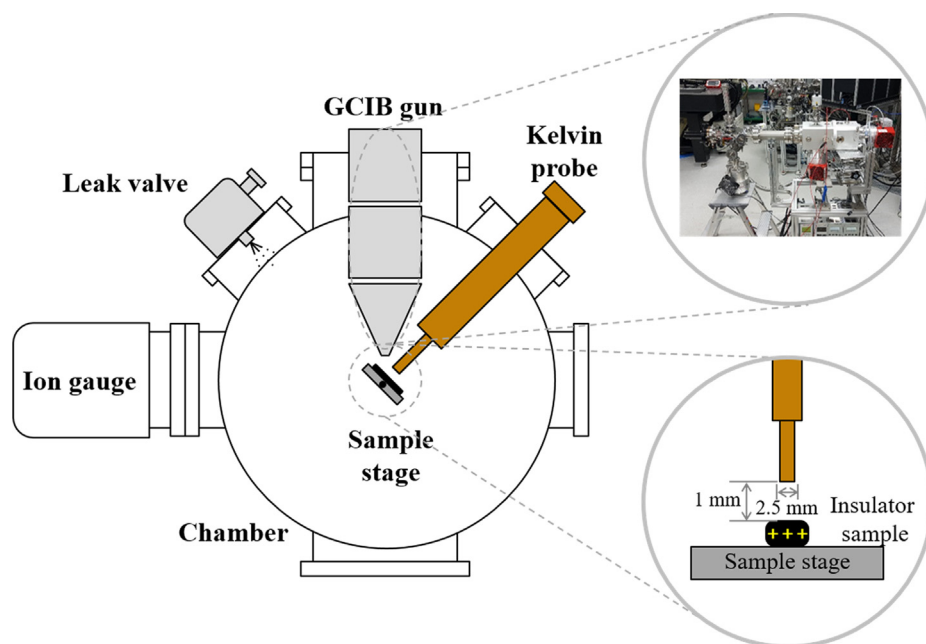


Fig. 1. Schematic diagram of the Kelvin probe measurement system consisting of the vacuum chamber, sample stage, GCIB gun, Kelvin probe, variable leak valve, and ion gauge.

The second part is the Kelvin probe device that measures the surface potential of the sample. The third part is the variable leak valve that changes the vacuum level in the chamber. The fourth parts are cold-cathode, hot-cathode ion gauges; and a vacuum chamber. The GCIB gun and the vacuum chamber are both pumped out using a turbo pump (260 L/s). The GCIB is the argon gas cluster ion gun (10 kV acceleration energy, 1,400 atoms/cluster, 10nA beam current) developed by KBSI. And it was installed at an angle of 45° with respect to the normal direction on the surface of the sample stage. The specific performance of the GCIB gun is provided in a previous paper [4]. A Kelvin probe (KP 6500, McAllister Technical Service, E Sherman Ave, USA) was placed in a port of the chamber perpendicular to the sample such that the probe tip surface is parallel to the sample. The probe tip surface has a length of 2.5 mm in diameter. The distance between the sample surface and tip surface of the probe was manually controlled using a linear translator installed at the Kelvin probe. The fine adjustments of the sample-to-tip distance and tip oscillation were made using a computer-controlled voice coil system. The Kelvin probe can be calibrated by using a reference direct current (DC) voltage applying on the sample stage over a range of 0 to 400 V (see Fig. S1).

The Kelvin probe measured the surface potential at 1-mm distance from the metal surface where the standard DC voltage is applied. As the applied standard DC voltage increases, the potential measured by the Kelvin probe increased. To check the potential signal variation by the chamber pressure, a standard DC voltage (200 V) was applied, and the chamber pressure was changed while the cold cathode ion gauge was turned on (see Fig. S2). When the pressure was gradually increased to 1×10^{-6} , 5×10^{-6} , and 1×10^{-5} torr, the average measured potentials were 171 V, 167 V, and 169 V, respectively. Even if the chamber pressure was changed, the Kelvin probe signal was not affected. To change the pressure in the chamber, a leak valve was mounted in the chamber at a distance of 170 mm from the sample surface. N_2 gas was injected through the valve. A cold-cathode ion gauge (PKR 251, Pfeiffer Vacuum, Asslar, Germany) and hot-cathode ion gauge (PBR 260, Pfeiffer Vacuum, Asslar, Germany) were alternately used in this chamber. Detailed diagrams of the cold and hot cathode vacuum gauges are shown in Fig. S3. In order to confirm the effect of distance between the ion gauge and the sample surface, distances of 130 and 400 mm were used alternately. The ion/electron current

generated in the vacuum ion gauge was measured with a source meter (KEITHLEY 6430 Sub-Femtoamp Remote SourceMeter, Keithley Instruments Inc., Cleveland, OH, USA) on the sample plate. Polyvinyl chloride (PVC) tape was used as a polymer sample by attaching it to a sample plate, and it was chosen due to its good adhesion and dielectric insulation properties [12].

3. Results and discussion

To understand the analysis conditions of the insulator samples in ToF-SIMS combined with GCIB, it is necessary to measure the surface potential formed on the polymer surface by the primary ion beam. For this measurement, a device involving a Kelvin probe was constructed, and the surface potential that was formed on the polymer surface was measured immediately after irradiation with the ion beam. The distance between the ion gauge and the sample was 130 mm. As shown in Fig. 2, the surface potential of the polymer measured by the Kelvin probe was

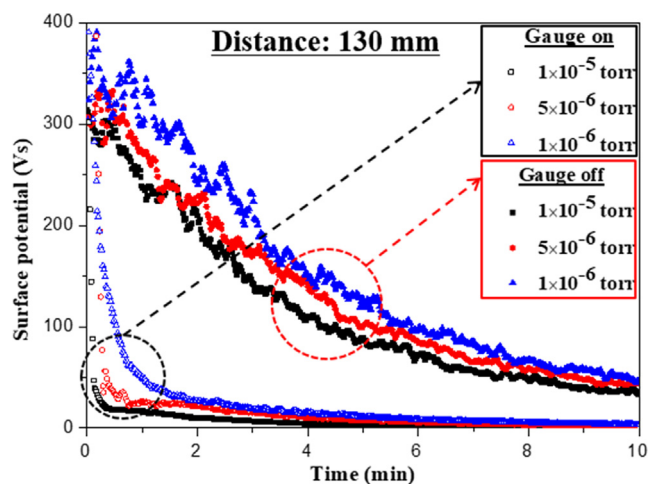


Fig. 2. Surface potential of the polymer film was measured by the Kelvin probe with the vacuum gauge either on or off condition at each chamber pressures adjusted on the distance of 130 mm between the cold cathode ion gauge and the sample plate following the irradiation of GCIB.

maintained for a relatively long time compared to the ToF measurement time (< 1 ms/cycle). The noisy shape of the potential value that appeared during the initial ~ 2 min as observed when the gauge was turned off can be attributed to the instability caused by the higher potential value compared to the distance between the Kelvin probe and the charged surface. This characteristic can be confirmed using the dispersion of calibration data presented in Fig. S1, and it can be confirmed that the measurement is stabilized at potentials below 150 V. From the measured temporal changes of the surface potential, it can be seen that the total potential of the surface rapidly decreased immediately after the irradiation of the ion beam. Fig. 2 shows the surface potential of the polymer film measured by the Kelvin probe at each differing chamber pressures following the irradiation of GCIB. The injection of N_2 gas adjusted the pressure of the chamber. It can be seen that there are different recovery times until the surface potential drops to zero depending on turning on or off the high vacuum gauge. As you can see, even when the vacuum gauge is turned off, the charge neutralization is relatively weak. However, it can be seen that when the vacuum gauge is turned on, it is mainly affected by the vacuum gauge. To confirm this observation, the current that reached the sample plate (grounded) was measured, as shown in Fig. 3 (a). When the cold cathode ion gauge is turned on, the current entering the sample plate achieves a value of a few nA, and this current disappears when the vacuum gauge is turned off. It can be seen that this current increase as the base pressure in the chamber increases. High vacuum gauges are able to ionize the neutral particles around the gauge and measure the current in these particles to determine the degree of vacuum. The above results show that this phenomenon inside of the high vacuum gauge can affect components far outside of the gauge. To examine the measured current in more detail, the current was measured with a DC voltage range of -20 V $\sim$ $+20$ V that was applied to the sample plate from a source meter. Both cold and hot cathode ion gauges were tested under these conditions. Fig. 3 (b) shows there is positive and negative charges inflow at constant chamber pressures after applying a DC voltage of -20 V to $+20$ V to the sample plate, which should be generated by a cold cathode ion gauge, while Fig. 3 (c) shows the same measurements with a hot cathode ion gauge. It was confirmed that the amount of current produced by the cold cathode ion gauge was relatively higher than that generated by the hot cathode ion gauge. Therefore, the subsequent surface charge experiment with the Kelvin probe was performed using the cold cathode ion gauge.

Additionally, in Fig. 3 (b, c), most of the current measured at positive DC voltages results from electrons, and the current from the cold cathode gauge of up to 35 nA was measured at a pressure of 5×10^{-5} Torr. The current measured at negative DC voltages represent the positive ion current, and as the pressure increases, the ion current increases. It can be seen that current with opposite polarity is arrived according to the polarity of the surface potential. And the magnitude of this current also depends on the degree of vacuum and the surface potential.

The results in Fig. 4 show that the cold cathode ion gauge directly affects the surface charge based on the pressure. It can be seen that the surface potential changed according to the degree of vacuum and the distance. As confirmed above, when the degree of vacuum increases and distance of the high vacuum gauge approaches, the surface potential decreases more rapidly. This effect can be used as a method of charge neutralization because it always serves to reduce the formed potential. However, the uncontrolled injection of electrons or ions can adversely affect the analysis of the sample.

To directly investigate the effect of surface charge reduction from different chamber pressures, Fig. S4 was normalized. Reduction times to the 1% amount of charge are plotted in Fig. 5 with a function of the pressure in the chamber. As the degree of vacuum increases, the reduction time of surface potential was increased rapidly. In this experiment, the Kelvin probe data was used to calculate the current flowing from the cold cathode ion gauge. First, the temporal changes of

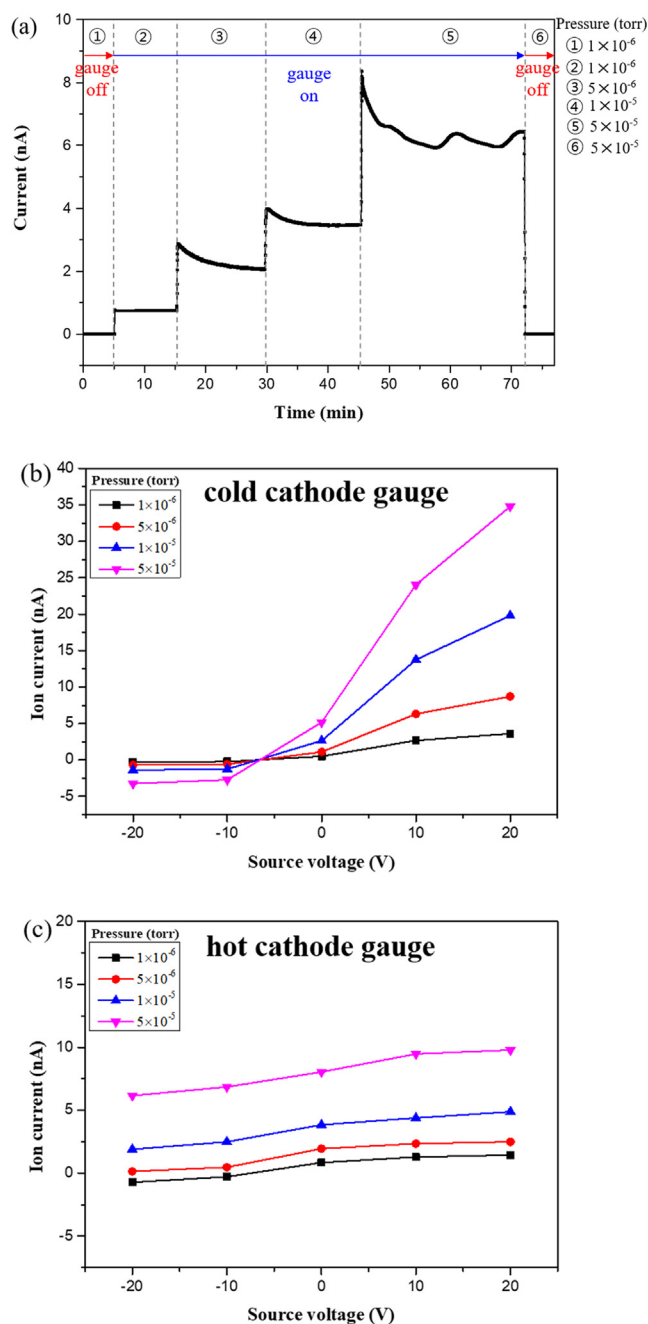


Fig. 3. Leaked current generated by the cold cathode and hot cathode gauge was measured at the distance of 130 mm between the high vacuum gauge and the sample. (a) Current generated by the cold cathode gauge at differing chamber pressures before and after the cold cathode gauge was turned on. (b) Current generated from the cold cathode gauge at differing chamber pressures after applying a source voltage of -20 V $\sim$ $+20$ V on the sample plate. (c) Current generated from the hot cathode gauge at differing chamber pressures after applying a source voltage of -20 V $\sim$ $+20$ V on the sample plate.

the surface potential data according to the vacuum degree were fitted to obtain fitting functions for each vacuum degree. Here, the fitted function represents the change of the surface potential with time. The surface charge, Q_s , and surface potential, V_s , are proportional to one another, and the proportional constant between them is the capacitance, C . The capacitance is determined according to the distance between the end of the Kelvin probe electrode and the charge accumulation area of the sample surface. And the charge accumulation area can be assumed as near a cross section area of the primary ion beam. This relation may

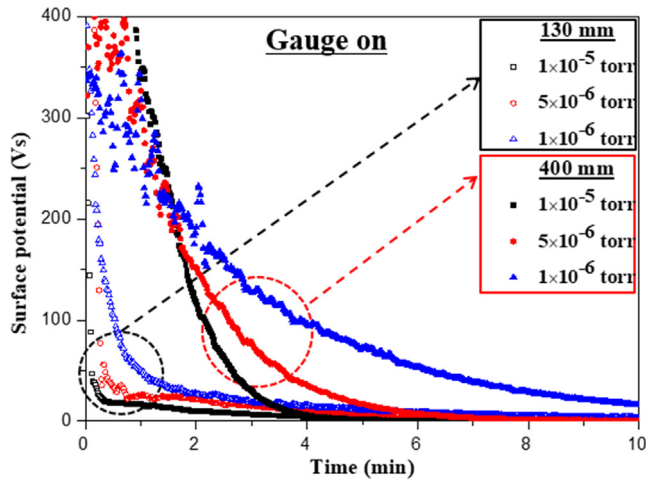


Fig. 4. Surface potential, V_s , of the polymer film measured at the distance of 130 mm and 400 mm between the cold cathode ion gauge and the sample.

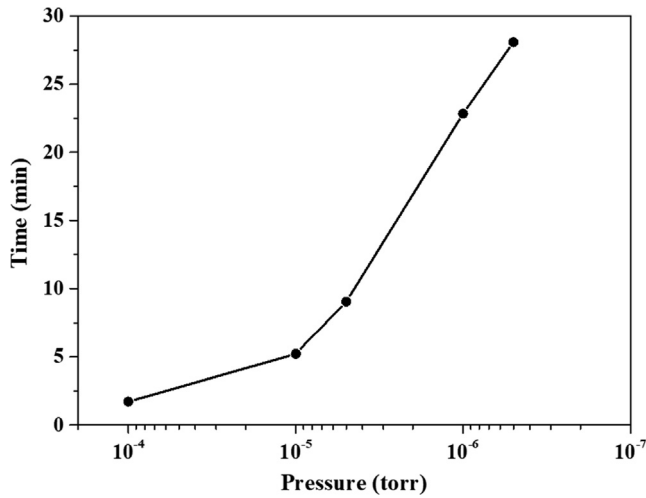


Fig. 5. The reduction time to the 1% amount of charge, which data was normalized with respect to the initial intensity of Fig. S4, according to the chamber pressure.

be expressed as follows:

$$Q_s(t) = CV_s(t) \quad (1)$$

$$C = \epsilon_0 \frac{A}{Z} \quad (2)$$

$$\frac{dQ_s(t)}{dt} = C \frac{dV_s(t)}{dt} \quad (\because \frac{dQ_s(t)}{dt} = I_s) \quad (3)$$

where $V_s(t)$ is the surface potential measured by the Kelvin probe and $Q_s(t)$ is the surface charge. C is the capacitance between the Kelvin probe and the sample surface, ϵ_0 is the vacuum permittivity, A is the cross-sectional area of the Kelvin probe ($\varnothing$ 2.5 mm), and Z is the distance (1 mm) between the Kelvin probe and the sample. If both sides of Eq. (1) are differentiated with respect to time, Eq. (3) is obtained and provides the total leakage current, I_s . This current is the sum of the total amount of leakage current through any path from the surface of the insulator (Or we can think it as a surface charge neutralization.). Included in this leakage current is the current flowing out through the surface and volume of the sample. The total leakage current also has an opposite polarity with regards to the current from the high vacuum gauge. Here, it can be assumed that the amount of current leaked to the volume and surface path [13] of the sample resulting from changes in the vacuum degree is almost unaffected. Therefore, the difference in the

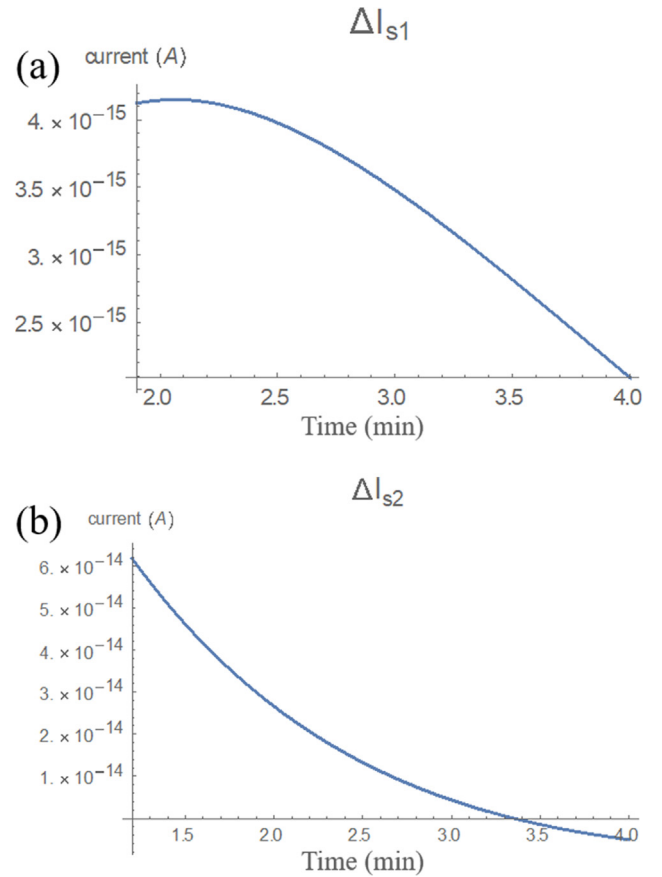


Fig. 6. Differences in the leakage currents at different vacuum degrees over time between (a) 1×10^{-6} torr and 5×10^{-7} torr and (b) 5×10^{-6} torr and 1×10^{-6} torr.

leakage current generated by the change of the vacuum degree may be represented the current difference injected from the vacuum gauge by the difference of vacuum level:

$$\Delta I_{s1} = I_{s(p=1 \times 10^{-6} \text{ torr})} - I_{s(p=5 \times 10^{-7} \text{ torr})} \quad (4)$$

$$\Delta I_{s2} = I_{s(p=5 \times 10^{-6} \text{ torr})} - I_{s(p=1 \times 10^{-6} \text{ torr})} \quad (5)$$

In the above equation, $I_{s(p=1 \times 10^{-6} \text{ torr})}$ is the differential function of the fitting function of the $V(t)$ data measured at a vacuum degree of 1×10^{-6} torr. Eq. (4) calculates the difference between the I_s values obtained at 1×10^{-6} torr and 5×10^{-7} torr, while Eq. (5) calculates this difference between the values obtained at 5×10^{-6} torr and 1×10^{-6} torr. In Eq. (4), the examined pressures differ by 5×10^{-7} torr.

In Fig. 6(a), it is shown that the current arrived from the gauge is approximately several fA at a pressure difference of 5×10^{-7} torr. A larger pressure difference of 5×10^{-6} torr is shown in Fig. 6(b), and this is almost ten times higher than the former pressure difference. The value of the gauge current injected by this pressure difference is several tens of fA, which is ten times higher than the amount calculated at the former pressure difference. This current from the high vacuum gauge may affect the analysis in TOF-SIMS.

4. Conclusions

We have constructed a device that uses a Kelvin probe and a gas cluster ion beam to induce and measure the surface potential on the surface of an insulator sample. Immediately after the gas cluster ion beam was irradiated, the value of the surface potential induced on the surface and its temporal change was measured. It was confirmed that the measured surface potential value is affected by the high vacuum

gauge. Also, there was an inflow of current with the opposite polarity of the surface potential from the vacuum gauge. And it was experimentally confirmed that the quantity of this current differs value depending on the degree of vacuum, and the position of the vacuum gauge. After the distance between the sample plate and the high vacuum gauge was set at a long, the current flowing from the high vacuum gauge was reduced. The current was calculated using the temporal change data of the surface potential measured at different vacuum degrees. At a pressure difference of 5×10^{-7} torr, the amount of current flowing into the insulator surface is on the order of approximately several fA. These high vacuum gauges were verified to exert small influences on the insulator analysis of ToF-SIMS. Improving the structure of these gauges such that they can be controlled appropriately may allow for this effect to be used as a charge neutralizer.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by a research grant from the Korea Basic Science Institute (D39613).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2020.146561>.

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